

DEVAYAN MANDAL

Full Stack AI Native / Forward Deployed AI Engineer & Open Source Contributor - Open to Remote Work

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PROFESSIONAL SUMMARY

Full Stack AI Native / Forward Deployed AI Engineer with 10 years of remote, self-directed production work. Built 30+ production-grade systems across autonomous multi-agent platforms, MCP integrations, deterministic RL evaluation, real-time observability, and high-throughput web applications. Merged 20 upstream pull requests across TypeScript, JavaScript, Python, and Rust in Recharts, pnpm, node-postgres, SQLMesh, Biome, Better Auth, thirdweb, Anza Kit, and Rspress. Specializes in explicit state machines, bounded autonomy, verifiable AI systems, hybrid RAG and memory architectures, streaming infrastructure, and production-grade developer tooling, with end-to-end ownership from architecture through deployment, observability, testing, and upstream delivery.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, R, Bash

AI & Agents: LangGraph, LangChain, LangServe, LangSmith, Model Context Protocol (MCP), multi-agent orchestration, supervisor routing, Reflexion loops, human-in-the-loop gates, hybrid RAG, RLVR environments, deterministic evaluation, prompt engineering, Ollama, Gemini, OpenAI-compatible providers

Frontend: React 19, Next.js 16, TanStack Start/Router/Query, Tailwind v4, Vite, Framer Motion, Expo, Three.js, React Three Fiber, HTML5 Canvas, SVG

Backend & Data: FastAPI, Node.js, Drizzle ORM, Prisma, SQLAlchemy, PostgreSQL (pgvector), Neon, Supabase, SQLite, Pydantic, Zod, Stripe

Realtime & Observability: SSE, WebSockets, Socket.io, replayable traces,

OpenTelemetry-shaped tracing, LangSmith evaluation pipelines

Cloud & DevOps: Vercel, AWS, Microsoft Azure, GitHub Actions CI/CD, serverless, scheduled pipelines, Vitest, pytest, Playwright, Biome

Web3 & Blockchain: Solidity, OpenZeppelin, Foundry, viem, wagmi, ethers.js, Safe Protocol Kit, RainbowKit, Reown AppKit, Base L2, blockchain and wallet infrastructure

Practices: Six Sigma DMAIC, bounded autonomy, retrieval-bounded generation, AST-sandboxed grading, cost-aware routing, threat-model documentation, Git, Changesets

PROFESSIONAL EXPERIENCE

LATTICE AI LABS

Full Stack AI Native / Forward Deployed AI Engineer | January 2016 - Present | Remote

- Architected and shipped 30+ production-grade systems across a broad portfolio of public engineering repositories spanning autonomous agents, RL evaluation environments, real-time observability, and high-throughput web applications.

- Built AEGIS, an enterprise AI ops cortex where a LangGraph supervisor routes 6 specialist agents across 7 agentic loops, with hybrid RAG, human-in-the-loop gates, LangSmith evaluation gates, and a live SSE dashboard. Live: aegis-agent-api.vercel.app/ui

- Built Synthesis, an autonomous research platform running cyclic Reflexion workflows (Plan, Research, Synthesize, Critique, Finalize) on a StateGraph executor with HITL approval gates, and AAROP, an inspectable 6-phase reasoning engine with cost-aware model routing, circuit breakers, RAG memory, replayable traces, and 99% Python test coverage. Live: synthesis-gold.vercel.app | aarop.vercel.app

- Built Veritas, a LangGraph research agent with SSE streaming, grounded demo mode, external retrieval when configured, and explicit separation between deterministic demo execution and live model/retrieval execution. Live: veritas-engine-woad.vercel.app

- Authored 3 official RLVR environments on the Prime Intellect Environments Hub (calendar-math, meeting-slot, unit-fix) using deterministic UTC sweep-line solvers and AST-sandboxed execution graders, removing LLM-as-judge variance.

- Implemented MCP servers and clients across Stdio and StreamableHTTP/SSE transports with client-side sampling and roots-based security boundaries; shipped HoldSlot, a WebMCP implementation for the OpenAI WebMCP Challenge 2026.

- Engineered Cadence, a framework-free healthcare agent with 5-layer clinical memory (session, episodic, semantic, knowledge, insights) using Ollama and SQLite, producing consent-scoped clinician briefs. Live: cadence-healthcare.vercel.app

- Shipped Jobrow, a live US tech job board indexing 5,000+ still-open roles across employer ATS boards (Greenhouse, Ashby, Lever) on TanStack Start, React 19, and Neon Postgres, with a public JSON API, Expo iOS/Android client, and GitHub Actions crawl pipeline running 4x daily. Live: jobrow.vercel.app
- Shipped Lattice, a live Web3 job platform for blockchain, crypto, and DeFi roles with same-day close detection from employer ATS boards. Live: lattice-devtechedge1.vercel.app
- Built a real-time chaos engineering platform with self-healing microservices, animated SVG topology, automated fault injection, and sub-second WebSocket telemetry, plus CollabSpace multiplayer whiteboards on Supabase Realtime; wired replayable OTel-shaped traces and LangSmith evaluation pipelines through the agent systems. Live: chaos-simulation.vercel.app
- Delivered Pulsar, a Base L2 GPU marketplace with Foundry smart contracts; RegTrace-AI, a MiCA/VARA compliance copilot with source-traced citations and hallucination flags; Smart Waitlist with Stripe-backed referral flows; and AstraMart.
- Operate a serverless daily AI news agent on GitHub Actions that pulls six public feeds, dedupes against in-repo memory, summarizes with Gemini, and delivers one Telegram executive brief at zero infrastructure cost.
- Work AI-native across the full stack using frontier coding and reasoning models through dedicated development harnesses, with emphasis on rapid iteration, test-driven validation, repository-specific contribution policies, and reproducible upstream delivery.

SPECTRAL QUANTITATIVE TRADING

Quantitative Crypto Trader & Derivatives Strategist | September 2016 - Present | Remote

- Manage diversified proprietary capital across tier-1 exchanges including Binance, Coinbase, Bitget, and MEXC, and on-chain protocols including Hyperliquid, Uniswap v3, Aave, and GMX.
- Built and ran automated grid, DCA, and Martingale bots on Pionex and Binance to capture spread and volatility in ranging markets.
- Executed delta-neutral basis and funding-rate arbitrage between spot and perpetuals, plus concentrated-liquidity and lending-loop strategies on-chain.
- Applied TWAP/VWAP execution on illiquid pairs, with VaR limits, drawdown caps, and Sharpe-ratio optimization governing position sizing and portfolio risk.

OPEN SOURCE CONTRIBUTIONS

Merged upstream:

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- [pytest-dev/pytest-env #262](https://github.com/pytest-dev/pytest-env/pull/262) (TypeScript) - document [pytest] env configuration in `pytest.toml` / `.pytest.toml` Merged Sep 2026. github.com/pytest-dev/pytest-env/pull/262
- [pnpm/pnpm #14863](https://github.com/pnpm/pnpm/pull/14863) (Rust/TypeScript) - fixed startup crashes on FreeBSD and other non-Windows Unix-like platforms by making `default_store_dir` use the Unix fallback path; added platform-specific regression coverage. Merged Sep 2026. github.com/pnpm/pnpm/pull/14863
- [web-infra-dev/rspress #3678](https://github.com/web-infra-dev/rspress/pull/3678) (TypeScript) - fixed a Search initialization race by awaiting `FlexSearch.addAsync` indexing before `LocalProvider.init()` resolves, preventing early searches from incorrectly returning no results on larger sites; focused regression-oriented fix. Merged Sep 2026. github.com/web-infra-dev/rspress/pull/3678
- [recharts/recharts #7805](https://github.com/recharts/recharts/pull/7805) (TypeScript) - removed `tabIndex=-1` from z-index portal SVG `<g>` layers so empty groups are not pointer-focusable and WebKit does not draw geometry-traced focus rings; added regression coverage asserting no layer carries a `tabindex` attribute. Merged Sep 2026. github.com/recharts/recharts/pull/7805
- [brianc/node-postgres #3772](https://github.com/brianc/node-postgres/pull/3772) (JavaScript) - stopped `Connection.sync()` from incorrectly setting the internal `_ending` flag, preventing false suppression of subsequent socket errors; added unit coverage for Sync vs disconnect semantics. Merged Sep 2026. github.com/brianc/node-postgres/pull/3772
- [pnpm/pnpm #14756](https://github.com/pnpm/pnpm/pull/14756) (Rust/TypeScript) - preserved existing dependency range operators and `npm:/jsr:` protocol prefixes during pnpm update, including prerelease and exact-version handling. Merged Sep 2026. github.com/pnpm/pnpm/pull/14756
- [pnpm/pnpm #14754](https://github.com/pnpm/pnpm/pull/14754) (Rust) - fixed non-recursive pattern runs with `--no-bail` so matching scripts continue executing and failures are aggregated correctly; added concurrent and sequential regression tests. Merged Sep 2026. github.com/pnpm/pnpm/pull/14754
- [pnpm/pnpm #14753](https://github.com/pnpm/pnpm/pull/14753) (Rust/TypeScript) - fixed `lockfile:false` being ignored when automatic package-manager switching downloaded the configured engine; added unit and filesystem-backed regression coverage across Rust and TypeScript paths. Merged Sep 2026. github.com/pnpm/pnpm/pull/14753

- SQLMesh/sqlmesh #6040 (Python) - moved ModelTest.create_test() off concurrent worker threads to eliminate a time_machine/ttl_cache race in sqlmesh test; verified the focused race regression repeatedly. Merged Sep 2026. github.com/SQLMesh/sqlmesh/pull/6040
- anza-xyz/kit #2032 (TypeScript) - restored number predicates in the getPatternMatchCodec advanced guide to match current codec typing. Merged Sep 2026. github.com/anza-xyz/kit/pull/2032
- better-auth/better-auth #11208 (TypeScript) - added regression coverage locking the /phone-number/verify OpenAPI requestBody after a Zod .and(z.record()) intersection; tests only, no production generator changes. Merged Sep 2026. github.com/better-auth/better-auth/pull/11208
- biomejs/biome #11667 (Rust) - ported the unicorn better-dom-traversing rule into Biome's nursery linter; +1,972 lines across 17 files and 7 commits, released to 13 packages via changeset. Merged Sep 2026. github.com/biomejs/biome/pull/11667
- thirdweb-dev/js #8938 (TypeScript) - fixed useTokenQuery error handling so 401/429/timeout failures rethrow instead of rendering a false "Token Not Supported" state in CheckoutWidget; added tests and a patch changeset. Merged Sep 2026. github.com/thirdweb-dev/js/pull/8938
- devtechedge/regulatory_compliance #4 (TypeScript) - allow localhost API in CSP connect-src for e2e Merged Sep 2026. github.com/devtechedge/regulatory_compliance/pull/4
- devtechedge/regulatory_compliance #3 (TypeScript) - restore full data/ seed tree for backend + e2e Merged Sep 2026. github.com/devtechedge/regulatory_compliance/pull/3
- devtechedge/veritas-engine #3 (TypeScript) - drop CSP that broke next-dev Playwright e2e Merged Sep 2026. github.com/devtechedge/veritas-engine/pull/3
- devtechedge/luxe-tracker #7 (TypeScript) - actually remove double comma after typescript block Merged Sep 2026. github.com/devtechedge/luxe-tracker/pull/7
- devtechedge/aarop #2 (TypeScript) - drop CSP that broke next-dev Playwright e2e Merged Sep 2026. github.com/devtechedge/aarop/pull/2
- devtechedge/notion-clone #2 (TypeScript) - convert next.config.ts to next.config.mjs for Next 14 Merged Sep 2026. github.com/devtechedge/notion-clone/pull/2
- devtechedge/regulatory_compliance #2 (TypeScript) - array.from for Set + restore data/frameworks seed Merged Sep 2026. github.com/devtechedge/regulatory_compliance/pull/2

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Active upstream engineering spans AI infrastructure, developer tooling, package management, frameworks, accessibility, concurrency, portability, security, SDKs, wallets, and blockchain infrastructure. Canonical ledger: github.com/devtechedge/oss-contributions

CERTIFICATIONS

- Model Context Protocol: Advanced Topics - Anthropic, 2026
- AI Fluency for Builders - Anthropic (with CodePath), 2026
- Agents and Workflows - OpenAI Academy, 2026
- Deep Agents with LangGraph - LangChain Academy, 2026
- Ambient Agents with LangGraph - LangChain Academy, 2026
- Microsoft Certified: Azure Fundamentals (AZ-900) - Microsoft, 2020
- AWS Fundamentals Specialization - Amazon Web Services (Coursera), 2020
- Data Science Specialization - Johns Hopkins University (Coursera), 2020
- Six Sigma Green Belt - Kennesaw State University (Coursera), 2020

EDUCATION

UNIVERSITY OF MUMBAI

Bachelor of Engineering (B.E.), Chemical Engineering (First Class) | [Graduation Year]
 Quantitative core in process dynamics, transport phenomena, mathematical modeling, and optimization; transitioned into software engineering through computational and systems-oriented work.